

# Adarsh Dwivedi

## Software Engineer

+91-9305597756 | [23ucs509@lnmiit.ac.in](mailto:23ucs509@lnmiit.ac.in) | [linkedin.com/in/adarshdwivedi30](https://www.linkedin.com/in/adarshdwivedi30) | [github.com/adarshcod30](https://github.com/adarshcod30)

Final-year CS engineer who ships, deploys and tests. Built an agentic platform on Node.js and MCP covered by 674 tests in CI, an analytics platform joining 59,985 police records into one queryable graph and deployed for a state force in 50 days, and a caching proxy on PyPI that fails open rather than failing the request. At Deloitte, migrated a production system onto new infrastructure without breaking existing frontend contracts.

## EXPERIENCE

### Deloitte India, Digital Excellence Centre

Jun 2026 - Aug 2026

AI Engineer (Agentic AI) Intern

Gurugram, India

- Built FastAPI and Pydantic backend services with schema-validated request and response contracts, powering tool-calling pipelines on an enterprise SAP-integrated platform.
- Root-caused a silent startup failure where an async Postgres driver deadlocked against Windows network I/O scheduling, evaluated every workaround, and selected the foundation the rest of the migration was built on.
- Authored the migration plan moving a production system onto new infrastructure without breaking existing frontend contracts, then validated both execution paths against a parity checklist before cutover.

## PROJECTS

### AGENTIQ, Agentic API Testing and Security Validation (B.Tech final-year project) | [Live](#) | [GitHub](#) Node.js, Express, MCP, Docker, Vitest

- Built an API testing and security platform on Node.js and MCP that turns a URL and plain-English intent into executable test cases with status, JSONPath, header and body assertions, grounded in an imported OpenAPI 3.1 document.
- Designed the tool layer as 19 Zod-validated tools across 6 risk classes with per-host grants, a filesystem jail and a process sandbox, and isolated state-mutating endpoints so a test run cannot corrupt the target.
- Covered the system with 674 tests in CI and tied test-generation confidence to the handler source rather than to the model's own assertion.

### Kadi, Crime Analytics Platform for the Karnataka State Police (KSP Datathon 2026) | [Live](#) |

TypeScript, Node.js, Zoho

[GitHub](#)

Catalyst

- Built and deployed an end-to-end analytics platform on Zoho Catalyst over 50 days, joining 59,985 FIRs across 31 districts and 298 stations into one queryable link graph with per-FIR ego networks, 5 layouts and 6 link-type filters.
- Shipped a command dashboard with disposal funnels, hour-by-weekday heatmaps and forecasts with 95% bands, plus evidence OCR and a grounded bilingual English and Kannada assistant that refuses to translate FIR numbers, dates or identifiers.
- Built one ranked worklist over 26,168 flagged cases, ordering by severity then urgency against the peer median, each row carrying a plain-language reason and a next action.

### CacheLLM, Semantic Caching Proxy for LLM APIs (Published on PyPI) | [Live](#) | [GitHub](#)

Python, FastAPI, Embeddings, Redis, PyPI

- Teams pay full price to answer the same question reworded. Built a drop-in proxy that keeps the OpenAI request and response shape including streaming, verified against the official Python and Node SDKs and LangChain, so adopting it costs one base-URL change.
- Routed six providers behind one interface from each host's live model list, with the cache as an in-process numpy matrix by default and Redis optional, so it runs with nothing else installed.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, JavaScript, TypeScript, SQL

**Frontend:** React, Next.js, TypeScript, Tailwind CSS, Vite, Recharts, MapLibre GL, Streamlit, Responsive UI

**Backend and APIs:** FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Redis, Zod Schema Validation

**Infrastructure and Tooling:** Docker, AWS, Google Cloud Run, Vercel, Render, GitHub Actions, CI/CD, Linux, Git, Observability, Reproducible Builds

**Core Computer Science:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

## ACHIEVEMENTS

- Adaptive Graph Search Suite: 12 routing algorithms in C++20 over 1.7M junctions, Contraction Hierarchies 44x faster than Dijkstra, verified exact against a reference implementation and compiled to WebAssembly. [GitHub](#) · [Live](#)
- Hacksplosion 2026, Deloitte India: cleared Levels 1, 2 and 3 with an AI-driven resume-to-job matching pipeline.
- 32 public repositories, most deployed and free to use, with every reported number reproducible from the repository and failures published beside results.

## EDUCATION

### The LNM Institute of Information Technology, Jaipur

2023 - 2027

B.Tech, Computer Science and Engineering

CGPA: 7.11 / 10

## LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.